

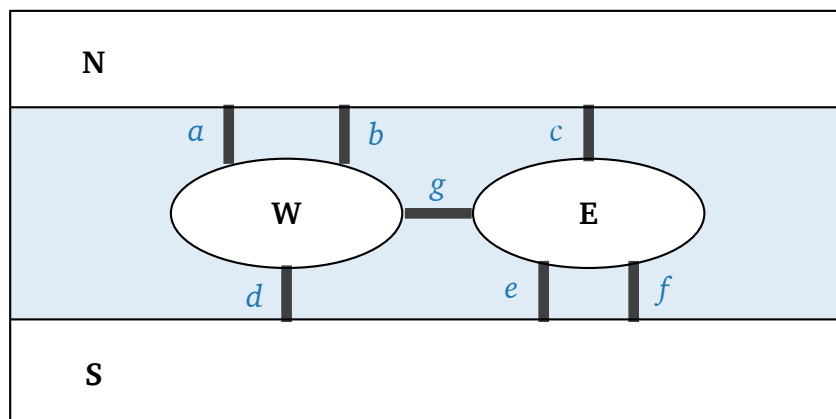
Cross Every Bridge Once, Then Teach a Computer to Do It

Work through this **before** the lecture. You are not supposed to know the technical words yet. Every word you need is defined on the sheet itself, and always *after* you have already built the thing it names. Use a pencil.

Part 1 — Seven bridges

A university campus sits on both banks of a creek. There are two islands in the middle. Seven footbridges, *a* to *g*, join the four pieces of land: the north bank **N**, the south bank **S**, the west island **W**, and the east island **E**.

You are a campus tour guide. Every footbridge has a donor's name on it, so your tour must cross **every bridge exactly once**.



Question 1(a): Find a route that crosses all seven bridges, each one exactly once. You may start on any piece of land and finish on any piece of land. You may step on the same piece of land more than once.

Write your route like this: land, bridge, land, bridge, ... The first two steps are done for you. Start on N, cross bridge *a* to reach W, then cross bridge *b* back to N.

N *a* W *b* N _____

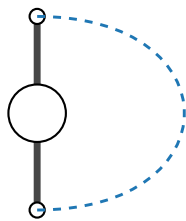
Question 1(b): Draw an **eighth** bridge on the map above. Call it *h*, and draw it between W and E, just below bridge *g*. The map now has eight bridges. Call it **Map B**.

Try again on Map B: cross all eight bridges, each one exactly once. Write your best attempt, and write which bridge was left over at the end.

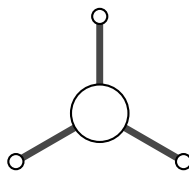
Question 2: Forget the whole map for a moment and look at **one** piece of land, with only the bridge-ends that touch it.

You are in the middle of your tour. Every time you *arrive* you use up one bridge, and every time you *leave* you use up another. So the bridges at this piece of land get used in **pairs**: one to come in, one to go out.

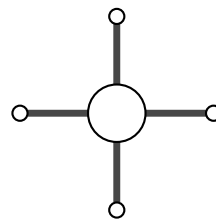
Below are four pieces of land, with 2, 3, 4 and 5 bridges. On each one, join two bridge-ends with a curved line, like this , to mean “in here, out there”. Keep making pairs until no more can be made. The first one is done for you.



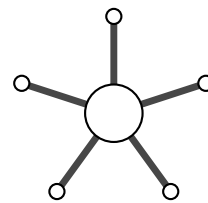
2 bridges



3 bridges



4 bridges



5 bridges

	2 bridges	3 bridges	4 bridges	5 bridges
pairs you drew	1			
bridges left over	0			

(a) On which of the four can you walk straight *through*, without ever starting or finishing there? _____

(b) What do those numbers of bridges have in common? _____

(c) A left-over bridge has no partner, so you can only use it once. That means this piece of land must be

- ☐ the start of the tour ☐ the finish ☐ the start or the finish

Question 3: Back to the map. Count the bridges at each piece of land, and write how many of the four have a **left-over** bridge. Map A, N is filled in: three bridges touch N, which is odd, so N has a left-over.

	N	W	S	E	how many have a left-over?
Map A (7 bridges)	3				
Map B (8 bridges)					

A tour has one start and one finish. So at most _____ pieces of land are allowed to have a left-over bridge.

Map A: ☐ a tour exists ☐ no tour

Map B: ☐ a tour exists ☐ no tour

Finish the sentence, using the words *left-over* and *start*:

No route can exist on Map B because _____

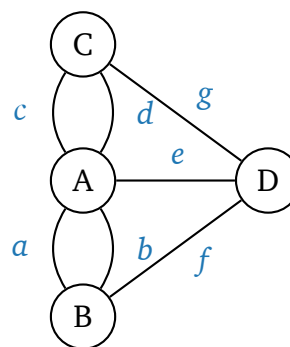
Question 4: Königsberg, 1736. A real city with four pieces of land A, B, C, D and seven bridges *a* to *g*. Two pairs of bridges join the same two pieces of land; both bridges of a pair must be crossed.

Count the bridges at each one:

A ____ B ____ C ____ D ____

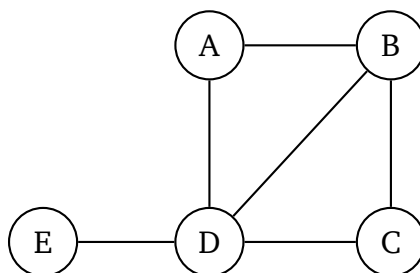
Use your rule from Question 3. Can the walk be done? ☐ yes ☐ no

In 1944 two bridges were destroyed and the walk became possible. Destroying just **one** bridge is already enough. List every single bridge that would work: _____



Part 2 — Three kinds of route

Forget bridges. Here is a different network: five bus stops, and the lines are direct bus services between them.



Question 5(a): For each route below, look at what it repeats. The first row is filled in for you.

route	repeats a line?	repeats a stop?	name
A B D B C	yes (B–D twice)	yes (B)	
A B D C B			
A B C D E			

Question 5(b): Here are the names. Write the right one in the last column of the table above.

A route that repeats **nothing** is a **path**.
 A route that may repeat stops but **never repeats a line** is a **trail**.
 A route that repeats **both** is just a **walk**.

Question 5(c): Write a route from A to E that repeats a stop but never repeats a line. (That is: a trail which is not a path.)

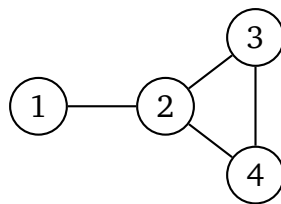
A _____ E

Question 5(d): Go back to your seven-bridge route in Question 1(a). Using these three names, it is a ☐ walk ☐ trail ☐ path.

Question 5(e): That route can never be a path. Finish the count that proves it: a path with 7 lines visits _____ different pieces of land, but the campus map only has _____.

Part 3 — Take these predictions to the laptop

A computer cannot look at a picture. Fill this page in **first**, on paper, then open the lab with your partner and find out if you were right. Here is a new network, with four nodes.



Question 6(a): Fill in the grid. Write **1** in row i , column j if a line joins node i and node j . Write **0** if it does not. Row 1 is done: node 1 is joined to node 2 only.

$A =$

	1	2	3	4
1	0	1	0	0
2				
3				
4				

Question 6(b): Add up each row of your grid, and compare each total with the number of lines meeting that node in the picture.

row totals: _____

What do you find?

Question 6(c): A **2-step route** from node i to node j means: one line from i to somewhere, then one line from there to j . Count them on the picture, by hand.

2-step routes from node 1 to node 3: how many? _____ list them: $1 \rightarrow _____ \rightarrow 3$

2-step routes from node 1 to node 2: how many? _____ why? _____

2-step routes from node 2 back to node 2: how many? _____

Question 6(d): In the lab you will multiply your grid by itself, giving a new grid called A^2 . Write down, now, what you expect to find in it.

I expect row 1, column 3 of A^2 to be _____ row 1, column 2 to be _____

and the diagonal (row 1 col 1, row 2 col 2, ...) to be _____

Now open the lab, in pairs, and bring this sheet — it asks for the numbers you just wrote down. One laptop between two people: the person who is *not* typing says what they expect before the other one runs the cell.

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